

### Advanced (Habanero)

**Q1.** Solve for  $x$  in the equation  $2^x = 8$

- (A)  $x = 2$
- (B)  $x = 3$
- (C)  $x = 4$
- (D)  $x = 5$
- (E)  $x = 6$

**Q2.** Solve for  $y$  in the equation  $3^y = 27$

- (A)  $y = 1$
- (B)  $y = 2$
- (C)  $y = 3$
- (D)  $y = 4$
- (E)  $y = 5$

**Q3.** Solve for  $z$  in the equation  $4^z = 16$

- (A)  $z = 1$
- (B)  $z = 2$
- (C)  $z = 3$
- (D)  $z = 4$
- (E)  $z = 5$

**Q4.** Solve for  $x$  in the equation  $x^2 = 25$

- (A)  $x = 3$
- (B)  $x = 4$
- (C)  $x = 5$
- (D)  $x = 6$
- (E)  $x = pm5$

**Q5.** Solve for  $y$  in the equation  $y^3 = 64$

- (A)  $y = 2$
- (B)  $y = 3$
- (C)  $y = 4$
- (D)  $y = 5$
- (E)  $y = 6$

**Q6.** Solve for  $z$  in the equation  $z^4 = 81$

- (A)  $z = 1$
- (B)  $z = 2$
- (C)  $z = 3$
- (D)  $z = pm3$
- (E)  $z = 4$

**Q7.** Solve for  $x$  in the equation  $2^{x+1} = 16$

- (A)  $x = 2$
- (B)  $x = 3$
- (C)  $x = 4$
- (D)  $x = 5$
- (E)  $x = 6$

**Q8.** Solve for  $y$  in the equation  $3^{y-1} = 9$

- (A)  $y = 1$
- (B)  $y = 2$
- (C)  $y = 3$
- (D)  $y = 4$
- (E)  $y = 5$

#### Open Ended Questions (FRQ)

**Q9.** Solve for  $x$  in the equation  $x^2 - 4x - 8 = 0$  using exponential equations

**Q10.** Solve for  $y$  in the equation  $2^y + 4^y = 80$

#### Chef's Tips

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- Tip 1: Make sure to simplify the expressions using exponent rules
- Tip 2: Pay close attention to the bases and exponents when solving equations
- Tip 3: Use the properties of exponents to rewrite equations in a simpler form